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# The contribution of wood-based construction materials for leveraging a low carbon building sector in europe

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ABSTRACT

Increasing the use of engineered wood products in the European Union can contribute to leveraging a shift towards a more emission-efficient production of construction materials. Engineered timber products have already been substituted for carbon and energy intensive concrete and steel-based building constructions, but they still lack the capacities and market demand to be more than just a niche market.

However, in the post-crisis period after 2008 the consumption of engineered wood products began rising in Europe. In this paper we analyse options for the future development of engineered wood products taking into consideration policy barriers and technical and environmental potentials for accelerating market introduction as part of a comprehensive scenario approach. For the European building sector we assessed an achievable potential for net carbon storage of about 46 million tonnes $CO_{2\text{-eqv.}}$ per year in 2030. To unlock this potential a bundle of instruments is necessary for increasing the market share for engineered wood products against the backdrop of existing policy instruments such as the gradual introduction of stricter rules for carbon emissions trading or more incentives for the voluntary use of innovative wood construction materials.

## 1. Introduction

The European building sector represents a significant material stock for wood-based construction materials.

In the 1950s and 1960s, 5.2 million tonnes of conventional wood-based construction materials (mainly sawn wood) were consumed on average every year (FAO, 1971). Between 1990 and 2010 this number was 6.4 million tonnes (UNECE/FAO 2008; United Nations 2010). The increased use of solid wood and timber panel constructions began to expand at the end of 1980s in the western European countries, such as the UK, Austria, Italy and Germany. In the last two decades the technical innovations of engineered timber products and their production processes, as well as the newly adopted building regulations e.g. fire protection regulations, have facilitated growth in the construction of multi-storey buildings made of wood. At the same time, the overall construction of and permits for new buildings decreased significantly after the financial crisis in 2007 (Eurostat, 2015; UNECE/FAO, 2013). In the decade from 2000 to 2010 the number of building permits were 1.7 to 2.5 times higher than in 2010 and even faced a further decrease of around 10% in the five years up until 2015 (Dol & Haffner 2010; Eurostat, 2015). A related market demand for wood-based construction materials, such as engineered wood products (EWP), in the building sector is expected for the construction of new residential buildings (one and two-family houses, multi-family houses and other types of dwellings) and for energetic renovation and retrofitting of old buildings, i.e. within urban built infrastructures (Weimar & Jochem, 2013). The construction of new residential buildings in the coming decades was therefore chosen as the main focus of the subsequent analysis. Considering the expected climate effectiveness of wood construction in residential buildings, both the retrofitting and the new constructions are regarded as a way to meet the European climate targets of decreasing $CO_2$ emissions by 88–91% by 2050 as indicated in the scenarios of Boermans, Bettgenhäuser, Offermann and Schimschar (2012). In fact, in terms of the total production volume of EWP, the ecological construction industries have been able to maintain constant growth rates of 2.5% to 15%, for example, in the production of wood fibre insulation boards (WFIB), cross-laminated timber products (CLT), laminated veneer lumber (LVL) and glulam products in recent decades. At the same time they have created added value by having a lower environmental impact and higher emissions efficiency than reference production systems (Hetemäki, 2014; Mahapatra & Gustavson, 2009; UNECE/FAO, 2013, Werner & Richter, 2007). A major strength of these

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**Table 1**
List of selected EWP included into the assessment (adapted from (Paulitsch & Barbu, 2015)).

| Products and associated NACE codes | Reference materials | Utilisation in structural components | Equality of benefits for comparative assessment | Development activities, market access and market volume | Production systems | Drivers for competitiveness |
|---|---|---|---|---|---|---|
| Oriented strand boards (OSB), medium-density fibre boards (MDF), wood fibre insulation boards (WFIB), (C1621) | EPS and rock-wool (C20.1.6) | ● Wood panel constructions<br>● Insulation | ● Overall heat transfer coefficient (W/m2*K) | ● Innovations: 1934–1990<br>● On the market since 1990 | Refiner process and thermal pressing processes | ● Material properties<br>● Policy incentives |
| LVL, glued laminated timber (Glulam) (C1621) | Steel beams and armouring steel (B7.1, C24 | ● Wood frame constructions<br>● e.g. beams | ● Compressive strength<br>● Bending moment (MN*m)<br>● Flexural and compressive strength | ● Innovations: 1971-today<br>● On the market since 2006 | Veneer peeling and press lines | ● Load bearing capacities<br>● Price performance ratio |
| Cross-laminated timber (CLT) (C1621) | Concrete (B8.1, 23.5), sand-lime bricks (B8.1, C23.6 | ● Solid timber constructions<br>● e.g. load-bearing walls | ● Flexural and compressive strength | ● Innovations: 1993–today<br>● On the market since 2006 | Saw mill and wood press | Multi-storey buildings/load bearing capacity, prefabricated products |

wood products is their highly optimised material efficiency and the central production of prefabricated products and turn-key projects which results in a cost-competitive construction time. Combining these innovations can open up a new space for opportunity for the wood construction industry in terms of more economically viable and energy efficient production (UNECE/FAO, 2013). The following product groups, with their respective NACE codes in brackets, are considered by the assessment to be the most relevant engineered wood products of the ecological construction industries: WFIB (C1621), LVL (C1621), Glulam (C1621) and CLT (C1621).

All of these materials are produced on an industrial scale and their market share of the construction sector is constantly increasing. Nevertheless, their market shares are still below 10% of total market demands for insulation materials and structural construction elements. The most dominant reference materials are compared in terms of their market shares and production-related emissions. In Table 1 the construction materials are characterised with regard to their functions, their product benefits, the drivers behind their increased market adoption and the historic facts regarding market access. The reference materials are also compared to wood-based products.

However, wood products used in construction applications remain part of a market niche in Central and Western Europe in particular because the absolute annual production volumes are still low compared to fossil-based and mineral-based construction materials (Mahapatra & Gustavson, 2009; Spirinckx et al., 2013).

Even though EWPs have superior technical properties, such as high bending moments, high flexural and compressive strength and time-efficient prefabrication, as well as ecological advantages such as low carbon intensity (Mahapatra & Gustavson, 2009), the fossil-based industries are more effective in playing out their competitive advantages. Assessing the question, if EWPs can play out their environmental advantages when a more level playing field is put into place, is a major motivation for implementing monitoring and assessment tools for the regional, for the national and the european context (Hildebrandt et al., 2017; Budzinski et al., 2017). The study presented in this paper focuses on the greenhouse gas (GHG) related impacts of the current and future use of EWP in the construction of residential buildings and on the policy drivers influencing the future adoption of EWP in the construction sector. It particularly aims to answer the following four research questions:

I What are plausible assumptions of future policy drivers influencing wood construction in residential buildings?
II What are the growth rates for the construction of new residential buildings and what are the recent annual consumption rates of EWP associated with this?
III What are the expected total GHG emissions and total carbon storage in building materials for the different future scenarios for using wood-based construction materials to build residential buildings in the EU27?
IV How do policy instruments affect future consumption rates of wood-based construction materials?

To answer these questions, the analytical framework of this study merges the assessment perspectives of i. the production of engineered wood products, ii. the analysis of policy drivers and iii. material flow scenarios for the consumption of wood-based construction materials.

This enables us to obtain a comprehensive picture of the potential contributions of the ecological construction industry for leveraging a low carbon building sector.

To do this, the production-related GHG emissions for the most dominant engineered wood products are assessed, taking into consideration their contribution to reducing GHG emissions compared to reference materials along past and future time series of wood product consumption in the construction sector.

The focus of the analysis is on the future use of engineered wood products and insulation materials with a particular focus on WFIB, CLT, LVL, OSB and glulam wood from an environmental and governance-related perspective.

The following section describes the scope of the analytical framework and methodological steps of the integrated assessment. It also provides an introduction to the applied methods. This is followed by a review of current policies that impact the ecological construction sector. The fourth section reviews the sustainability criteria and indicators of the ecological construction sector. A discussion on the drivers behind the expansion of capacity and how they can be steered is presented in Section 4 before the paper concludes in Section 5.

## 2. Analytical framework and methods

### *2.1. Analytical framework*

A context-specific analytical framework was developed in order to establish a methodological procedure for analysing the impact of policy instruments on the use of ecological construction materials in the construction of residential buildings. The analytical framework encompasses the following three system perspectives which are then addressed by the analysis:

- From **Governance perspective:** an outline of the relevant European policy instruments and their impact in different country-specific contexts.
- From a **market perspective:** the development of scenario steps supported by (i) a qualitative analysis of the impact of energy and climate policies on the construction sector and (ii) a trend analysis for the development of the European building sector for residential buildings.
- From the **perspective of material flows and GHG emissions:** a modelling of the material flows for the use of wood to construct residential buildings according to the data obtained from a literature review and a collection of statistical data and most plausible development trends. This enables the most relevant techno-economic parameters to be identified and the relevant development drivers to be specified for the use of engineered timber products within the wood construction sector.

Fig. 1 presents the three perspectives (policy instruments, wood construction market and material flows) and introduces the basic influencing factors of the analytical framework. The analytical framework serves as a template for capturing all of the relevant aspects for the analysis. It shows the extent to which existing measures potentially enhance the development of the construction industry and which instruments appear most suitable.

#### *2.1.1. Research approach and structure of the paper*

In order to set up an integrated methodology, the policy instruments are first assessed in terms of their influence on the construction material market. Then the factors describing and determining wood use in the construction of residential buildings are compiled and evaluated. The methodological approaches used to develop the scenarios for assessing the impact of the future use of EWP in the construction of residential buildings were carried-out in the sequence as presented under the five steps a) to e). Furthermore the expected results are described in the context of the overall assessment approach.

a) Characterisation of assumptions for policy drivers influencing or inhibiting the increased use of wooden construction materials in the overall building sector
b) Collection of data to specify historical developments and the status quo of floor space for residential buildings and wood resource flows in building constructions
c) Calibration and parametrisation of the material flow models for describing the status quo of material flows of engineered wood products for the construction of residential buildings and for assessing the GHG emissions of engineered wood products and selected reference construction materials from a life cycle perspective (Section 3)
d) Development of material flow scenarios for the future consumption of EWP incorporating the future development of the residential building sector based on the quantitative characterisation of the policy drivers behind the development trends most relevant for the construction sector (Section 4)
e) Assessment of the material flow scenarios of future consumption of EWP and the associated GHG accounting results (Section 4)

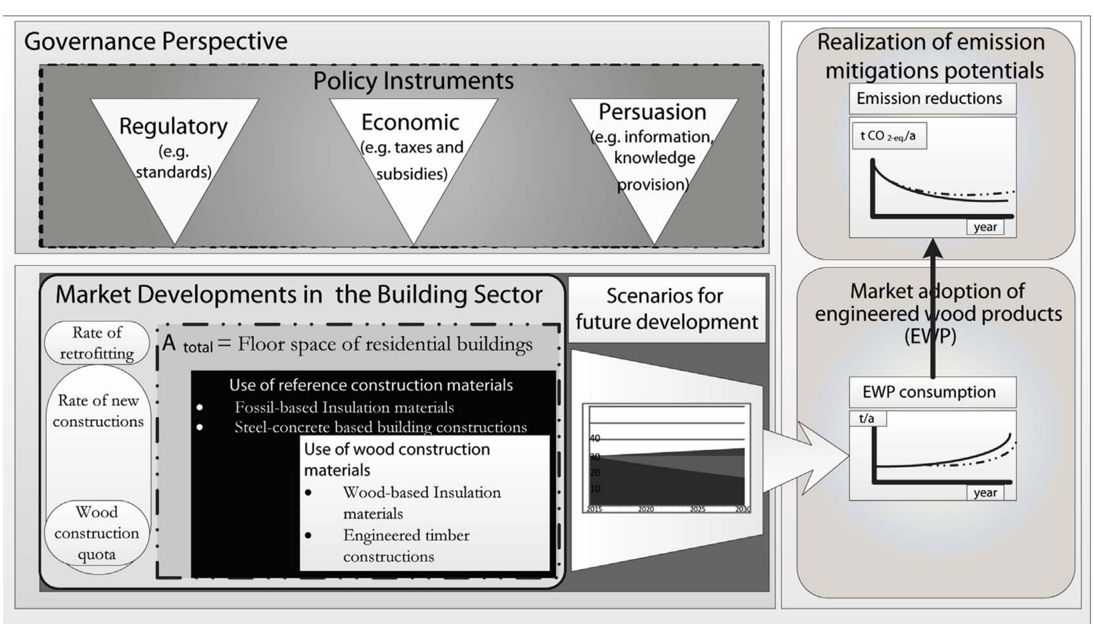


**Fig. 1.** Analytical framework of the study.

#### 2.1.2. Expected results

- Imaging scenarios for the future development of the building sector, the market for engineered wood products and the related GHG effects
- Evaluation of the potential influencing factors of policy instruments on the increased future use of ecological construction materials and assessment and discussion of the future potential emission reductions

### 2.2. Overview of material and methods

#### 2.2.1. Review of the policy instruments

The policy instruments relevant for wood product applications implemented in the national and European context play a key role in the future development of wood-based production capacities.

These instruments can be categorised in various ways depending on different factors such as detail degree (see Sterner & Coria, 2012 or Vedung, 2010 for an overview). However, they all aim at affecting behaviour by setting either positive or negative incentives (Van Tongeren, 2008). For this analysis we follow Vedung (2010) and apply three instrument categories: regulatory, economic and information (hereinafter called persuasive) instruments. Vedung (2010) alternatively calls them "sticks", "carrots" and "sermons": Regulatory instruments ("sticks") are binding rules and regulations that have to be followed. If they are neglected fines or sanctions are issued. The uptake of economic instruments ("carrots") is voluntary and they either offer or take away resources (Vedung, 2010). Persuasive instruments ("sermons") rely on influencing behaviour through information transfer. They are also called "soft" instruments because they are not binding and do not apply pressure. All three instruments differ mainly in terms of liability, the way they assert influence (coercion, pressure or voluntariness) and enforcement and monitoring costs. Conducting a comprehensive review of all existing policy instruments at European level was not feasible in the context of this paper. The reason is that language constraints and the fact that measures on wood construction are hard to identify because they are often only one out of many aspects within a policy.

Therefore the scenario analysis a literature review was conducted to identify i) selected potentially relevant policy sinstruments on national and international scales that influence the scenario assumptions (see Table 2) and ii) prominent and best practice policy instruments for the discussion in Section 4 on how to increase the proportion of wood construction and retrofitting across different European countries.

#### 2.2.2. Qualitative scenario analysis of policy interventions influencing the wood construction quota

For the scenario analysis, a normative situation was first developed which characterised the most dominant policy drivers influencing the stagnation of growth of the wood construction quota. The normative approach is conducted by developing a scenario frame to characterise the most dominant policy drivers influencing the stagnation or growth of the wood construction quota. The timeframe of the scenario covers the next 15 years until 2030. This time span was chosen in accordance with an analysis of the developments in the wood construction industries in the last 20 years from 1990–2010 and 2000–2011. A comparison of these developments shows diverging trends for the different wood products. The two additional scenarios developed for the comparison are based on the assumption that market demand and policies are the significant drivers of the wood-based building sectors.

Due to changes in prices or changing consumer preferences market demand for wooden building products could increase. However, for some products, the markets do not yet exist or are only niche markets because the processes and products are still in their infancy. "New technologies cannot immediately compete on the market against established technologies." (Geels & Schot, 2010: p. 80). Policies can

**Table 2**
Identified policy drivers influencing the future rates of retrofitting, new constructions and wood construction.

| | Scenario 1: Market driven development of wood construction | Scenario 2: Policy interventions to increase the proportion of wood constructions | Choice of policy Instruments | Specification of instruments |
|---|---|---|---|---|
| Scenario assumptions | Extrapolation of current trends | Sensitivity to policy interventions | | Country-specific policies and pan-European standards and directives |
| Rate of new constructions of residential buildings | ⬂ | ⬀ | Economic instruments | Interest rate policy, economic recovery programmes, state support for private pension provision |
| | | ⇨ | Regulatory instruments | Land use policies |
| | | ⇨ | Persuasive instruments | Pension provision |
| Rate of retrofitting of residential buildings | ⬀ | ⬀ | Economic instruments | Loan incentives, public-private funds, |
| | | ⬀ | Regulatory instruments | German Renewable Energy Sources Act |
| | | ⬀ | Persuasive instruments | Information campaigns |
| Wood construction quota for residential buildings | ⬀ | ⬀ | Economic instruments | Loan incentives, green procurement |
| | | ⬀ | Regulatory instruments | Binding targets for renewable materials share |
| | | ⬀ | Persuasive instruments | Labelling, information campaigns |
| Rate of demolition of residential buildings | ⇨ | ⇨ | Economic instruments | Circular economy, interest rate policy, subsidies |
| | | ⇨ | Regulatory instruments | Waste Management Act, building laws (structural engineering), the Soil Protection and Contaminated Sites Ordinance, protection of historic monuments |
| | | ⇨ | Persuasive instruments | Urban policy, neighbourhood policy |

support the development of markets by using instruments that differ with regard to the incentives they offer. Effective policy drivers could be economic instruments, such as subsidies for R & D or tax deductions on sustainable processes and products. Command and control instruments such as prohibiting the use of certain substances or by-products also have the potential to increase the uptake of wooden building materials. However, all policies have to be assessed carefully to avoid that that they remain piecemeal that neglect interactions and spillovers at different political and administrative levels.

#### *2.2.3. Scenario development and quantification*

In defining a future scenario for the most plausible development of wood construction quotas by 2030, construction quotas and trade balances in the northern European exporting countries are expected to remain high at around 75–85%. In contrast, the trend towards more domestic consumption in trading countries such as Austria, Germany, Italy, and France will slightly increase from around 5–10% to 15–20%. For both scenarios it is assumed that, according to the actual rates of new constructions in the European building sector, the average rate in the coming 15 years will still be low compared to the developments in the 1990s up until 2005 and will basically be market driven. The stronger growth in the UK and Germany will not significantly raise the total rate within Europe. Table 2 illustrates how policy instruments that were identified as having a potential impact on rates of new construction, retrofitting and new construction quotas could impact the direction of change in rates and quotas and provides examples of policies related to these instruments.

### *2.3. Modelling material stocks and flows of construction materials*

#### *2.3.1. Review of existing studies for data collection and for justification of the innovation*

A comprehensive literature review of existing studies for material flow analysis of construction wood was conducted in order to gather information on methodological approaches and to compile material flow data for scenario modelling. The majority of the studies address a) the development trends in the construction of residential buildings, b) the general quantification of the material intensity in the construction sectors, c) the categorisation and the quantification of mass fractions of wood-based construction materials for residential and non-residential buildings and d) the quantification of waste wood fractions within construction and demolition waste.

As presented in the materials and methods section, the most relevant factors are A = the area of residential floor space and its country-specific growth rate r, the wood construction quota $f_{wcq}$ and its time-related development, and the proportion of wood-based construction materials used to construct different building types and different age classes of buildings. Due to the major market disruptions induced in the construction sector by the financial crisis and the debt crisis, the analysis has to rely mostly on recent data but also to account for major market changes which have become evident in the last decade. These trends in the post crisis recovery of the building sector need to be thoroughly evaluated before they can be used for accurate projections of future material flow scenarios.

Therefore, the studies reviewed cover the time period from 1995 to 2009 and from 2009 to 2014. Studies, such as the results from the iNSPiRe project and from A. Müller, offer a thorough analysis of the building types and material stocks in the different ages classes of residential buildings (see Table 3).

There are a few studies that aggregate the wood construction metrics for residential buildings within a national context, for example, Mantau, Glasenapp and Döring (2015) for Germany, or for the Nordic countries. Despite remaining within the national context, most studies and policy papers represent only rough estimates of GHG emission reduction potentials and are not based on transparent material flow assessments and construction market projections (Jorma et al., 2010;

**Table 3**
Relevant studies on residential building stocks and material stocks of wood-based construction materials.

| Topic | Useful parameters | Authors (Date of Publication) | System boundary |
|---|---|---|---|
| Development of a model for prognosis of demolition waste as part of the material flow analysis for the circular economy | Mass fractions of waste wood in demolition waste of wooden and non-wooden buildings | Görg, H. Schriftenreihe WAR 98, Darmstadt, 1997 | German building sector |
| Economic development of the building sector and future market chances for marketing constructional wood | Mass fractions for different types of constructional wood in the construction of wooden and non-wooden buildings | Mantau & Kaiser, 2013 | German building sector |
| Tools for Management of Construction and Demolition Waste | Reference values for mass fractions of wood in buildings in Europe and Germany | Müller, 2014 | German and European situation for demolition waste |
| Europe's buildings under the microscope | Country by country distribution of residential floor space | Economidou et al., 2011 | European building sector |
| iNSPiRe-Development of Systemic Packages for Deep Energy Renovation of Residential and Tertiary Buildings including Envelope and Systems | EU-27–Residential stock, building types and construction | Birchall et al., 2014 | European building stock |

**Table 4**
Model parameters and eligible sources for model parameterisation.

| Parameter | Description | Parametrisation and data source |
|---|---|---|
| $A_{t,Residential}$ | Total area of residential floor space within Europe | (Economidou et al., 2011) and related study BPIE, 2011; (Birchall et al., 2014) and related project iNSPiRe 2014; Dol and Haffner (2010) |
| $A_{t,Residential\text{-}wood}$ | Residential floor space of wood constructions within selected EU countries | Calculated with $A_{t,Residential}$ and with Mantau et al. (2015); Bund Deutscher Zimmermeister (2014); Destatis (2016); Fleury & Chiche, 2006; ; UK Structural Timber Association (2013); UK Department for Communities and Local Government (2015); Centro Studi Federlegno Arredo Eventi SpA, (2015); Istat (2016); La Asociación de Fabricantes y Constructores de Casas de Madera de España (AFCCM) (2010 & 2015) |
| $\dot{m}_{EWP}$ | Consumption of engineered wood products in the construction of residential buildings | UNECE/FAO (2001, 2008, 2011, 2012, 2013, 2015); Höglmeier et al. (2013); IAB/Müller (2014); United Nations (2010). |
| $f_{\_(GHG\text{-}Emissions)}$ | Emission factors for engineered wood products | Ruuska (2013); Verband der Deutschen Holzwerkstoffindustrie e.V. (2013); Studiengemeinschaft Holzleimbau e.V. (2013); Hildebrandt et al. (2017) |
| $C_{\_\ (totalEWPGHG\text{-}Emission)}$ | Cumulated GHG emissions associated with all assessed EWP products groups | Calculated using Eq. (5) |

Mahapatra & Gustavson, 2009; Mantau et al., 2013).

### 2.3.2. Calculating the future material flows

The static material stocks in the sector for the construction of residential buildings is modelled

i by quantifying the floor space area $A_{total}$ of residential buildings in Europe (million $m^2$),
ii by specifying the structure of the residential buildings (% of single family houses, % of multi-family houses)
iii by quantifying the shares of construction materials in different types of structures (e.g. kg steel/$m^3$ in reinforced concrete structures) and the wood construction quota $f_{wcq}$
iv by specifying the ratio factors for the surface area to the gross building volume ($m^2/m^3$)
v by identifying the weighted average of the share of wood construction materials in the gross building volume of residential buildings (e.g. t EWP/$m^3$) (factor called $f_{GBV}$)
vi by quantifying the annual consumption $\dot{m}_{specificmaterial}$ of selected construction materials ($m^3$/a or t/a)

The proposed approach of Heeren, Jakob, Martius, Gross, and Wallbaum (2013) is modified in order to model the floor space area of residential buildings over time.

Eq. (1) for calculating floor space area of residential buildings and Eq. (2) for calculating share of wood constructions

$$A_{t,Residential} = A_{0,Residential} + \sum_{k=1}^{t} A_{k-1,Residential} \times r_{new,k,Residential} - \sum_{k=1}^{t} A_{k-1,Residential} \times r_{dem,k,c} \tag{1}$$

$$A_{t,Residential-wood} = A_{t,Residential} - A_{t,Residential} \times f_{wcq} \tag{2}$$

The time-related changes in the proportion of wood constructions and the overall floor space in residential buildings for future scenarios are modelled using:

- $r_{new}$: growth rates for the new construction of buildings
- $r_{dem}$: growth rates for the demolition rate
- $f_{wcq}$ = wood construction quota

Eqs. (3) and (4) for calculating material flows in the consumption of construction materials in the building sector

$$\dot{m}_{specificmaterial}[t] = n_{HouseType} \times A[m^2] \times f_{GBV}\left[{}^{m^3}\!/_{m^2}\right] \times n_{GBV-Material}\left[{}^{t}\!/_{m^3}\right] \tag{3}$$

$$\dot{m}_{totalEWP} = \sum_{MG1=m}^{n} \dot{m}_{MG1} + \sum_{MG2=m}^{n} \dot{m}_{MG2} \tag{4}$$

whereby MG 1 = material group of insulation and non-structural construction materials such as WFIB, OSB and MDF and MG2 = material group of structural material such as beams and stanchions, e.g. LVL, Glulam and CLT

### 2.3.3. Greenhouse gas accounting for the consumption of construction materials and their gradual substitution

Greenhouse gas accounting requires

A an analysis of sector-specific carbon emissions of production plants for various materials from different material groups (MG) and
B compilation and analysis of GHG data from life cycle assessments available from environmental product declarations (EPD) and from LCA benchmarking studies for these construction materials.

Eq. (5) for calculating material specific GHG emissions and Eq. (6) for calculating total GHG emissions of EWP:

$$C_{\text{total EWP GHG}-\text{Emission}}\left[kg_{CO_2-eqv.}\right] = \sum_{MG1=m}^{n} C_{MG1} + \sum_{MG2=m}^{n} C_{MG2} \tag{5}$$

$$C_{\text{Avoided GHG}-\text{Emission}} = C_{\text{total REF GHG}-\text{Emission}}\left[kg_{CO_2-total}\right] - C_{\text{total EWP GHG}-\text{Emission}}\left[kg_{CO_2-total}\right] \tag{6}$$

Remark: If C (avoided GHG emissions) is > 0, then substituting steel and mineral-based construction materials with engineered wood products offers the potential for reducing GHG emissions.

### 2.3.4. Overview of model parameters and sources for model parameterisation

The model parameters, as described in 2.3.3, are compiled in Table 4. The most relevant parameterisation sources are given for each parameter. The extensive data for the country–specific wood construction quota and for the product-specific emission factors was collected from national statistic agencies, industrial associations and databases for environmental product declarations. The most relevant sources for these two model parameters are also presented in Table 4.

### 2.3.5. Background data for the consumption of wood-based construction materials in the last decade

There were significant increases in the market share of timber frame constructions, wood panel constructions and solid wood constructions in several European countries between 2000 and 2010 (Enquête nationale de la construction bois, 2013; Fleury & Chiche, 2006; Mantau et al., 2015; UK Structural Timber Association, 2013). In France and the UK the wood construction quota grew by almost 9% in nine years. The

**Table 5**
Growth rates in the consumption of engineered wood products during selected time periods.

| Engineered wood products for timber constructions | Average annual growth rates in consumption from 2000–2014 | Growth rates from 2011–2014 | Sources |
|---|---|---|---|
| Sawn wood | −0.80% | 0.73% | UNECE/FAO (2001, 2008, 2011, 2012, 2013, 2015); United Nations (2010) |
| Wood fibre insulation boards (WFIB) | 12.17% | 1.25% | Ibid. |
| Medium-density fibre boards (MDF) | 0.90% | 1.25% | Ibid. |
| Cross-laminated timber (CLT) | 24.17% | 13.33% | Ibid. |
| Glued laminated timber (Glulam) | 8.57% | −0.60% | Ibid. |
| Oriented strand board (OSB) | 14.06% | 5.76% | Ibid. |
| Average annual consumption of wood-based construction materials [in 1000 t/a] | 15516.1 | 19239.7 | Own calculation |
| Weighted average growth rate related to the overall consumption | 3.09% | 1.96% | Own calculation |

most significant growth rates for timber products were achieved for cross-laminated timber and for Glulam timber. However, in terms of absolute percentages, sawn wood consumption remained the primary material used in constructing residential buildings. The background data used to calibrate the modelling of the consumption of EWP distinguishes between the growth rates during the post-crisis period and pre-crisis period. The growth rates for the specific EWPs for these two periods are presented in Table 5.

Fig. 2 provides an overview of the annual volume of EWP consumed in the building sector for wood constructions. These annual material flows are used to calibrate the growth rates and for modelling the future material flows consumed in the wood construction sector.

### *2.3.6. Wood construction quota in the context of general developments in the European building sector*

The area of heated floor space in the EU 27 is approximately $25{\cdot}10^9$ m$^2$. Useful floor space of residential buildings (64% single family houses and 36% apartment buildings) accounts for around $18.75{\cdot}10^9$ m$^2$ (75%) (Economidou et al., 2011). The cumulated floor space for Germany, France, Italy, Spain and the UK is around $12.4{\cdot}10^9$ m$^2$.

The wood construction quota has to be set in relation to the total number of building permits. In the post-crisis period after 2007, the total number of building permits and amount of building completions within Europe faced a significant downturn of around 50% (Eurostat, 2015; UNECE/FAO, 2013). Only in the UK and Germany was the construction sector able to slowly recover. The demolition rate within the European countries varies substantially and ranges from about 0.05–0.23% (Meijer, Itard, & Sunikka-Blank, 2009). Traditionally, the wood construction quota is high in northern European countries, which are also all net exporters of wood products with a strongly optimised value added chains within the wood industry (Mahapatra & Gustavson, 2009; Wall, 2013). Countries that are more trade oriented often have a lower wood construction quota whereas there is a more diverse situation in importing countries. Scotland, for example, has a very high quota, whereas the Netherlands has a low wood construction quota (Mahapatra & Gustavson, 2009; Wall, 2013). Overall it can be said that the production and consumption of wood products is locally concentrated in most of the countries with a high production capacity (OECD, 2015a, 2015b). This also explains why the wood construction quota within individual countries also varies significantly between different regions. In Germany, the wood construction quota in 2012 was highest in the southwest (Rhineland Palatinate and Baden Württemberg) at 21.6% and 23.7% respectively. In France the highest proportion of wood constructions used to construct new buildings was in the central east at 19% (Bund Deutscher Zimmermeister, 2014; Enquête nationale de la construction bois, 2013). All of these regions have a high forest density and a long tradition of wooden buildings. In addition to customer preferences and a strong tradition of a wood-based industry, the availability of mineral and metal construction materials plays a crucial role in the regionally specific construction types. A large proportion of European countries can be regarded as “self-sufficient” and are able to meet their “domestic demand” for construction minerals. The largest producers of construction minerals are France, Spain, Italy and Germany, which are all countries with a relatively low wood construction quota of 5% to 15% (Bahn-Walkowiak, Bleischwitz, Distelkamp, & Meyer, 2012). In order to compare the situation in the five major European countries and the rest of the EU 27, Table 6 presents the wood construction quota based on country, and recent annual numbers for the completion of residential buildings.

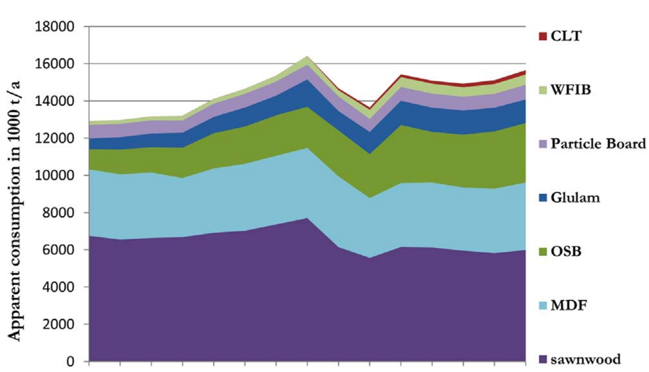


**Fig. 2.** Apparent consumption of wood-based products for the construction of residential houses (modified from UNECE/FAO, 2001, 2008, 2011, 2012, 2015).

## 3. Results

This section characterises, specifies and analyses the scenarios for future developments in the building sector, the market for engineered wood products, and the related GHG effects.

### *3.1. Driver quantification for future trends*

Based on a qualitative scenario analysis, the quantitative factors such as rate of retrofitting, rate of new constructions, rate of demolition and the wood construction quota are considered to be the drivers behind the two scenarios.

The following key assumptions were considered when scenario modelling the development of the building sector:

- Rate of retrofitting: 2%
- Rate of new constructions: 0.87% (The average rate of new constructions in relation to the total residential floor space was 1.2% in the period before the financial crisis and dropped in the post-crisis period to 0.8% (Dol & Haffner, 2010; Santonico et al., 2008)). The relative wood construction quota withstood the crisis and increased in several European countries, such as the UK, France and Germany,

**Table 6**
Country-specific characteristics in the construction of residential buildings (wooden and non-wooden).

| Parameters | Unit | European Countries | | | | | |
|---|---|---|---|---|---|---|---|
| | | Germany | France | UK | Italy | Spain | Others |
| Annual completion of useful floor space for residential buildings | [1000 $m^2$] | 17200 | 41000 | 14100 | 3800 | 2540 | 51360 |
| Wood construction quota | [related to completed residential buildings] | 15 | 12 | 25 | 6 | 6.5 | 4.5 |
| Average useful floor space per housing unit completed | [in $m^2$] | 113.6 | 99 | 83 | 73.5 | 130 | 108 |
| Annual completion of residential buildings | [number of dwellings] | 150833 | 345118 | 170690 | 51200 | 17000 | 475556 |
| Annual completion of wood-based buildings | [number of dwellings] | 16969 | 17083 | 17602 | 3072 | 456 | 8828 |
| Annual completion of wood-based buildings | [in 1000 $m^2$] | 1928 | 1691 | 1456 | 226 | 59 | 953 |
| which have wood frame structures | % of completed housing units | 50 | 74 | 92 | 55 | | |
| which have X-Lam structures | % of completed housing units | | | | 38 | | |
| Sources | | 1 Mantau et al., 2015; Weimar & Jochem, 2013; Destatis, 2016; Dol & Haffner, 2010 | 2 Fleury & Chiche, 2006; Enquête nationale de la construction bois, 2013; CEI BOIS, 2015; Euroconstruct, 2008; Dol & Haffner, 2010 | 3 UK structural Timber Association, 2013; UK Department for Communities and Local Government, 2015; Dol & Haffner, 2010 | 3 Centro Studi Federlegno Arredo Eventi SpA, 2015; Istat, 2016; Dol & Haffner, 2010 | 4 López Letón, 2015; Dol & Haffner, 2010; Ministerio de Fomento, 2016; | Dol & Haffner, 2010 |

**Table 7**
Quantitative characterisation of the impact of economic and policy drivers on construction rates.

| | Scenario 1: Market driven development of wood construction | Scenario 2: Policy interventions to increase the proportion of wood constructions | Derivation and substantiation of assumptions |
|---|---|---|---|
| Rate of new constructions (related to total area of residential floor space) | 0.87% | Average 1.0% (rising from 0.85% in 2015 to 1.15% in 2030) | Depending on the economic cycles, the rate of new constructions could increase with economic recovery<br>The economic dynamics in European countries are still fragmented (economic recovery in one European country does not necessarily reflect the situation in other countries).<br>Rate of construction depends on national policies (such as financial support) |
| Wood construction quota (related to total area of residential floor space) | 4.9% | Average 9.1% (rising from 4.9 in 2015 to 14% in 2030) | High path dependencies in regional product adoption (different in Spain than in the Scandinavian countries)<br>Wood resource consumption regionalised<br>In countries with urban settlement structures and citizen movements, steel-concrete and sandstone bricks remain the primary building materials |
| Rate of demolition (related to total area of residential floor space) | 0.175% | 0.175% | |

making up a significant share of total building permits (Enquête nationale de la construction bois, 2013; Mantau et al., 2015).

- Rate of demolition: 0.175%
- The wood construction quota was estimated to be around 5% on average with the potential to further increase if the timber market shares continue to expand in the construction sector in countries such as France and the UK.

The key assumptions for the scenarios were derived by analysing the developments of building permits, as well as the start and completion of building work on dwellings in the different European countries. Then the values for the annual completion of useful floor space in housing units were taken as the basis for calculating the development of floor space area for residential housing, the market share of timber constructions and the material flows and stocks of wood products. In Table 7 the main assumptions of the scenario modelling are presented and substantiated according to the findings of the qualitative analysis of future policy drivers.

### *3.2. Future trends in the construction of residential dwellings and future material flows of construction materials*

The annual completion of useful floor space for residential dwellings varied between 180 million $m^2$/a and 250 million $m^2$/a between 1992 and 2010 (Dol & Haffner, 2010; Eurostat, 2015). For the scenario modelling, a slight recovery in annual housing completions is projected (see Fig. 3). However, the maximum annual increase is estimated to be < 200 million $m^2$/a by 2030. The number of building permits decreased significantly between 2009 and 2014 by approx. 60%. (Eurostat, 2015).

For the time period between 2000 and 2009 the average annual growth rate of residential floor space was 1.3%, whereas it dropped to 0.85% in the years between 2009 and 2014. Therefore, the maximum future growth rate is estimated to range between 0.87% and 1.0%. The annual increase in useful floor space associated with new housing completions between 2015 and 2020 is expected to be below 150 million $m^2$/a which is the most realistic description of the current situation in the European construction sector.

The growth rates for the consumption of EWP are presented in Table 8 in order to make comparisons between the two scenarios. The average annual consumption of EWP for the period under investigation ranges between approx. 23.7 million tonnes and 30.9 million tonnes depending on the scenario-specific assumptions.

The weighted average growth rate of consumption of wood products for construction was 3.1%, including sawn wood, wood fibre insulation boards, medium density fibre boards, oriented strand boards, glulam timber and cross-laminated timber, with an annual consumption of 15 million tonnes per year for the construction of residential houses. This is significantly higher than the average growth rate for overall construction for the period up until 2009, but only slightly higher than the

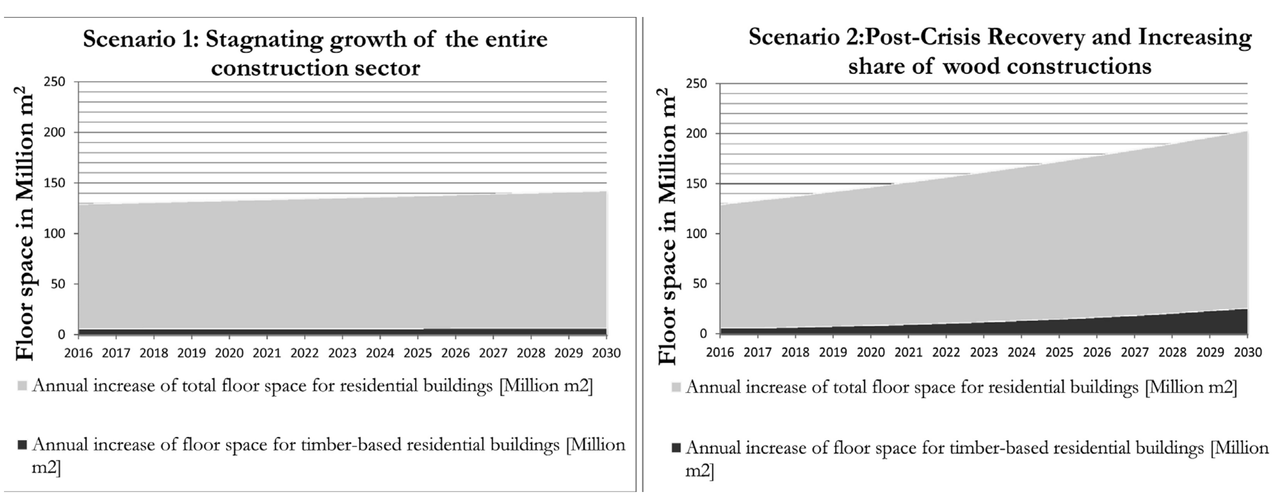


**Fig. 3.** Scenario trends for modelling future growth of residential floor space.

**Table 8**
Growth rates in the consumption of EWP for constructing residential buildings.

| Engineered wood products for timber constructions | Scenario 1: Assumed growth rates for 2015–2030 | Scenario 2: Assumed growth rates for 2015–2030 | Sources |
|---|---|---|---|
| Sawn wood | 1.88% | 1.88% | Own modelling results |
| Wood fibre insulation boards (WFIB) | 9.00% | 15.00% | Own modelling results |
| Medium-density fibre boards (MDF) | 1.50% | 3.50% | Own modelling results |
| Cross-laminated timber (CLT) | 13.75% | 26.50% | Own modelling results |
| Glued laminated timber (Glulam) | 9.00% | 14.70% | Own modelling results |
| Oriented strand board (OSB) | 5.80% | 7.50% | Own modelling results |
| Average annual consumption of wood-based construction materials [in 1000 t/a] | 23707.33 | 30933.81 | Own calculations |
| Weighted average growth rate related to overall consumption | 5.49% | 13.36% | Own calculations |

growth rate for the post-crisis period. Therefore, it is assumed that the total share of wood-based construction materials will increase in the post crisis period. The individual growth rates for cross-laminated timber (24.17%), LVL and Glulam products – three major EWPs – are characterised by a slight recovery towards pre-crisis growth rates for Scenario 1 and with a strong recovery in the expansion of production capacities and consumption rates for Scenario 2. Due to the short time series of some of the EWP growth rates the propagation of uncertainty levels is important to further interpret the robustness of the two scenario paths.

### 3.3. Potential future GHG emissions savings attainable by increasing utilisation of EWPs in the construction market by 2030

The product-related GHG emissions and the attainable emission savings were calculated based on the expected market developments of timber products used in the construction of residential houses as presented in the previous sections. Emission savings are calculated by using a material-specific substitution factor that compares wood-based construction materials to fossil-based comparators, thereby calculating the future potentials for substituting conventional construction materials such as EPS, stone wool, sand-limestone and reinforced concrete elements.

For the entire scenario modelling approach the individual levels of uncertainty were compiled and the combined uncertainty of all modelling parameters was calculated (see Table 9). For the exponential growth rates in the equation the propagation of uncertainties was included by applying a regression analysis. For all other parameters the uncertainty level was considered to be constant along the scenario paths. The combined uncertainty is slightly higher toward the upper boundaries and slightly lower toward the lower boundaries due to the circumstance that for the future projections already the emission factors from carbon efficient production processes were chosen. For the other factors except the carbon emission factor the uncertainty is equally distributed between the upper and lower boundaries. The overall uncertainty is used to show the possible ranges of the two scenario paths (Fig. 4). On the following diagrams the results for the two scenarios and their respective uncertainty levels are presented.

In Scenario 1 the cumulated annual GHG emissions increase from

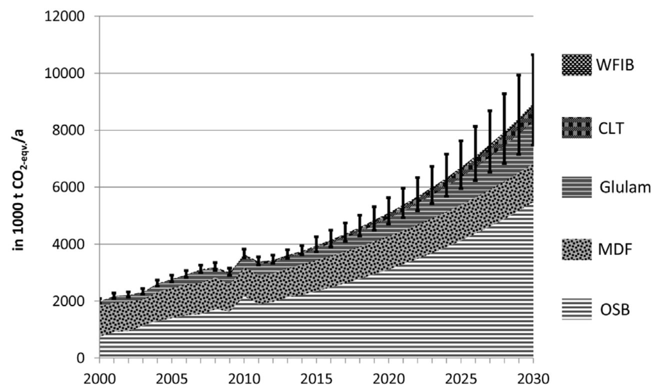


**Fig. 4.** Scenario 1 Annual GHG emissions for producing wood-based construction materials until 2030.

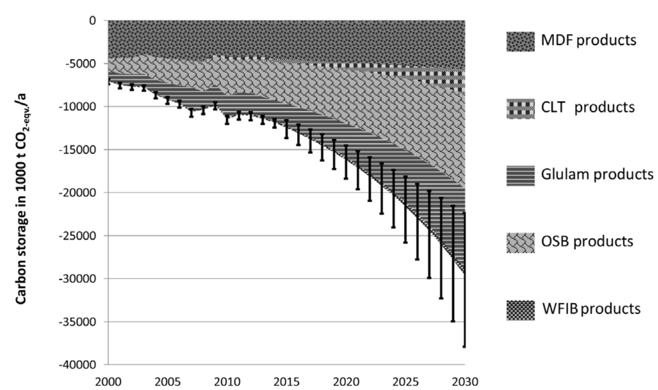


**Fig. 5.** Scenario 1 Carbon storage in the building stock by using wood-based construction materials in residential buildings.

approx. 3950 kt $CO_2$-eqv./a in 2015 to 8900 kt $CO_2$-eqv./a in 2030. The GHG emissions associated with the production and use of OSB represent the largest share by far at approx. 5500 kt $CO_2$-eqv./a in 2030. At the same time, the OSB-related carbon storage also has the highest share in total carbon storage for the building stock through the use of wood-

**Table 9**
Uncertainty analysis for interpreting the robustness and possible variances of the projected modelling results.

| Uncertainty factors | Uncertainty | Data category | affected modelling parameters | |
|---|---|---|---|---|
| Input uncertainty in construction statistics | ± 1.40% | Annual data points | area and relative consumption | $m^2$ and t/$m^2$ |
| Input uncertainty in density of wood products | ± 1.25% | Conversion factors and future material efficiencies | absolute mass flows in consumption | $m^3{}_{EWP}$ and $\rho = t_{EWP}/m^3_{EWP}$ |
| Variance in market shares due to uncertainty in dominating construction types | ± 2.50% | Annual data points | Relative market shares and relative consumption rate | t/a, $t_{EWP}/m^2$ and $t_{EWP}/m^3_{ofbuildingelements}$ |
| Input uncertainty in carbon emission factors | +4.00% −1.00% | Future benchmarks | Carbon emission factor | t $CO_{2\text{-eqv}}/t_{EWP}$ |
| Combined uncertainty of modell parameters | ± 5.08% (26%/46%) −3.28% | Including regression analysis | Total annual GHG emissions | t $CO_{2\text{-eqv}}$/a |

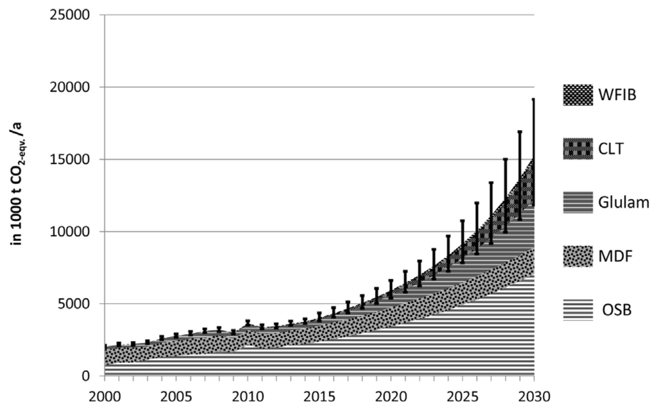


**Fig. 6.** Scenario 2 Annual GHG emissions related to consumption of wood-based construction materials until 2030.

based construction materials (see Fig. 5). While the carbon storage associated with the use of MDF remains stable, it still represents a significant amount of the full storage potential.

A comparison of the scenario results reveals that the OSB-related GHG emissions make up the highest share for both scenarios. Even though a strong growth rate was projected in Scenario 2 for other EWPs, such as Glulam and CLT, these other EWPs contribute less to the overall consumption-related GHG emissions.

In total, CLT and Glulam combined would contribute to a global warming potential of 1968 kt $CO_2$-eqv./a in 2030 (Fig. 6).

In contrast to the results of Scenario 1, the carbon storage associated with the consumption of CLT in Scenario 2 in the construction of CLT-based solid timber constructions becomes far more important and accounts for approx. 15,600 kt of $CO_2$-eqv. of storage potential in the building stock (Fig. 7).

## 4. Discussion and conclusions

In the scenario analysis we project that the annual demand for EWP materials up to 2030 is expected to range between 23,707 kilotons per year in Scenario 1 and 30,933 kilotons per year in Scenario 2. The related carbon accounting for the two scenarios indicates that the CLT-based solid wood structures and the glulam-based frame structures not only exhibit better material efficiency in terms of the total number of residential houses built, the increased use of these products would also result in lower annual GHG emissions and a higher carbon storage in the building stock possibly ranging between 29,650 up to 60,500 kilotons $CO_2$-eqv. The higher amounts of EWP volumes and GHG emission reductions can be realised only with a significant annual growth rates (i.e. 24% for CLT and 12% Glulam), what demands an high and constant growth in annual production capacities. The potential emissions savings in completion of residential floor space compared to conventional building materials are in a range between approx. 18,000 kilotons $CO_2$-eqv. in 2030 in Scenario 1 and approx. 46,000 kilotons $CO_2$-eqv. in Scenario 2.

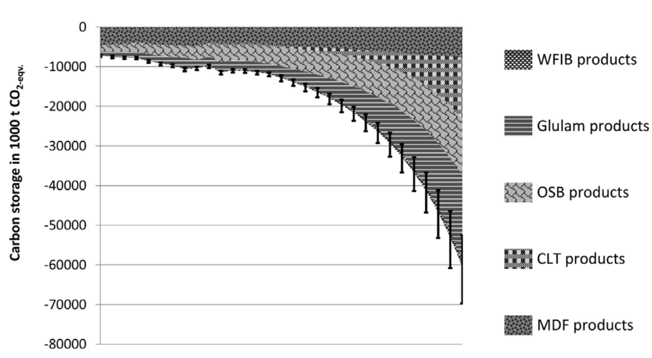


**Fig. 7.** Scenario 2 Carbon storage in the building stock by using wood-based construction materials in residential buildings.

Three development trends can potentially lead to a higher share of wood construction in the future: (a.) technical progress towards high performing load-bearing wood constructions, (b.) progress in the harmonisation of fire protection and environmental regulations, (c.) constant growth in the capacity to produce ecological construction materials (Östman & Källsner, 2010; UNECE/FAO, 2013).

This trend can in part be supported by tailor-made policies as demanded by several different policy strategies. At the global and European level, policies to enhance green innovations are requested, e.g. by “The Future We Want” (UN, 2012), the Roadmap to a Resource Efficient Europe (EC, 2011a) or the Roadmap for moving to a competitive low carbon economy by 2050 (EC, 2011b) and others (see for example EEA, 2014). The selection of the “right” instrument depends on the policy’s target (reducing $CO_2$ emissions, selecting easy to implement options, low cost alternatives or a combination of all three).

For our analysis we include policy instruments that can directly or indirectly influence rates of new construction or retrofitting, wood construction quotas for residential building or the rate of demolition by i) increasing costs for production or consumption of environmentally harmful products and ii) enhancing wood construction processes and products as well as supporting the wood construction resource base (see also Fig. 8). Because wood construction measures vary across Europe, the focus of the analysis is more on policy instruments and less on specific policies. We conclude that a bundle of measures is necessary to unlock the potential for a more sustainable building stock in the next decade.

### 4.1. Indirect wood construction policies

The promotion of fiscal reforms, further development of environmental taxes as well as emission trading schemes that can trigger “the resource-efficient green economy transformation process” are all instruments that restrict carbon-intense products and production processes and, as such, support ecological construction products (see Fig. 8). The phasing-out of subsidies that support fossil resources (EEA, 2014: p. 7) can also foster a sustainable building sector. Effective schemes to reduce greenhouse gas emissions could lead to an increase in timber construction and in the proportion of wood used in retrofitting because their price would relatively decrease. Taxes can either be designed to reduce the tax burden of companies that produce in an environmental friendly way or increase the tax burden of companies that increase environmental damage. However, the effect of the carbon pricing schemes to reduce carbon emissions is debatable because the increasing efficiency in the manufacturing sector is mainly driven by energy costs and less by policies (Morris, 2014). Moreover, under the current legislation in Europe, the cement sector – a high carbon emission sector- will not have to buy carbon certificates until 2043 and, as such, the amount of carbon produced by the sector will probably not decrease (Morris, 2014).

### 4.2. Direct wood construction policies

The uptake of innovative wooden construction components depends not only on market incentives but also on the mental models towards wooden products in the construction sector. These are an “internal representation that individual cognitive systems create to interpret the environment and the institutions are the external (to the mind) mechanisms individuals create to structure and order the environment” (Denzau & North, 1994: p. 4). For wooden products (Rametsteiner, Oberwimmer, & Gschwandtl, 2007: p. 24) show that “people generally regard wood as less fire resistant, less durable, less dimensionally stable, less resistant to decay and insects, and more expensive to maintain, than other materials.” The perception differs between using

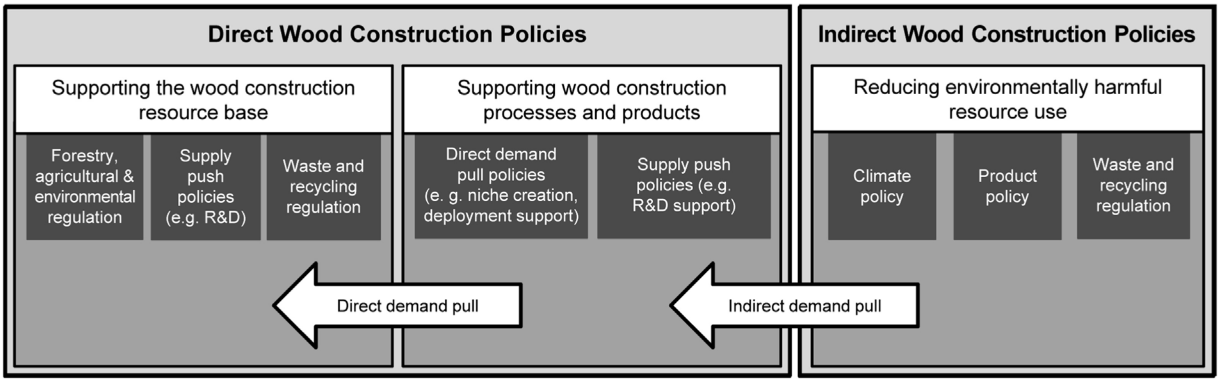


**Fig. 8.** Categorisation of wood construction policies (following Pannicke et al., 2015).

wooden material as a construction material where users are more cautious in terms of reliability and quality of the products compared with materials for renovation and re-modelling activities (Rametsteiner et al., 2007). Experiences with building standards and tradition have a great influence on the willingness to take up alternative materials such as wood components, one example being wooden window frames (Gustavsson et al., 2006). Consequently it will most likely influence the demand for products such as wood-based panels, glulam timber or CLT in Germany. The UK and France that already have high wood construction rates, but the Mediterranean, where the construction quota has stagnated since the economic crisis, do not. Here subsidies and persuasion are more suitable instruments for consumers to become familiar with these new products. A differentiation can be made between two different forms of information provision: Knowledge transfer from producers and politics to consumers through labelling or information campaigns, and between producers through patents. For example, so-called green patents are an indicator of the increase in green technologies. The spread of knowledge at an international level could increase with claims for patent protection at an international level however this is a cost-intensive and lengthy process (EEA, 2014). To effectively reduce quality and sustainability concerns regarding the increased use of wood, the demands by different groups, such as architects, designers and building owners, for the certification of wood from sustainable sources (Falk, 2010) has to be addressed by encouraging reliable and international certification schemes. However, certification often includes compatibility with strict fire regulation standards (Gustavsson et al., 2006) and structural inspections. To ensure sustainable wood production, the European Union Timber Regulation from March 2013 aims at preventing illegally logged wood in the EU (UNECE/FAO, 2013).

Education can also support the uptake of wooden building material, especially if planners and architects are taught about it at universities and provide spill-over effects to firms (Aschhoff & Sofka, 2009). Education and knowledge transfer about wood use in construction is key to overcoming path dependencies which are especially strong in the building sector where the market is dominated by stone materials (Goverse, Hekkert, Groenewegen, Worrell, & Smits, 2001). R & D support for innovations is a key element of a more sustainable economy because resource efficiency gains in Europe are mainly driven by changing technology (EEA, 2014). This support can be public and private "through innovative financing instruments, such as revolving funds, preferential interest rates, guarantee schemes, risk-sharing facilities and blending mechanisms" (EC, 2011b: p. 11). The IEA (2008) illustrates an example of demand-side subsidies in the Austrian state of Vorarlberg where all edifices receiving public subsidies have to be passive houses. Moreover, "[t]he increased use of passive houses has made these technologies widely known by constructors and users in Austria." (IEA, 2008, P. 70) Despite the impact of the financial crisis, procurement is another way to increase demand, such as through public funds to private companies.

Cultures and traditions, the political and economic systems but also the availability of wood influence the path a low carbon building sector will evolve in different European countries. However, to reach the optimistic growth rates for EWP (of 24%) for CLT, and 12.5% for WFIB and OSB in Scenario 2 regulatory instruments can place powerful incentives on the rate of wood construction quotas and the rate of retrofitting. The French example shows the effect of legal requirements on the admixture of wood materials in construction material. Restrictions on the use of certain mineral resources for construction processes would have a similar affect but their uptake is not realistic, especially in mineral-rich countries in the Mediterranean region. Additionally, more stringent regulations on the reduction of environmentally harmful substances used in the production processes could place sustainable resources in a more competitive position.

## 5. Conclusion

The investment decisions of a low carbon building sector have a long-term impact on the carbon emissions of the building environment and can serve in the short and medium term as an engine for accelerating innovation for producing construction materials within the industries. The potential emission reductions through the promotion of wood materials in construction industries have yet to be realized due to persistent market and policy barriers (Jorma et al., 2010). More precisely, the adoption is impeded "by a number of barriers including the resilience of the current technological, and social system, lack of knowledge about applications, lack of financing, insufficient incentives for replacing old technology with new, high costs, and insufficient market demand." (EEA, 2014: p. 43). It is especially difficult for SMEs to tap into the financial resources required to invest in new technologies for innovative processes and products. To answer the question of how to achieve a higher proportion of wooden building material in construction and retrofitting, a mix of policy instruments adapted to the specific economic, political and cultural features of the country, especially in terms of wood materials, is required. An example how to increase the proportion of wood construction is provided by a Swiss study (Amstalden et al., 2007) which showed that individual policy measures (subsidies, income tax deductions and carbon tax) had already impacted retrofitting activities, but that a significant effect was achieved by mixing these policy instruments.

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